



Predictive Modeling of Root-Zone Variables Using Greenhouse Data

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01 Motivation

Root-zone pH controls nutrient availability; EC (electrical conductivity) reflects salinity and fertilizer load.

Manual lab samples are accurate but sparse, creating blind gaps between measurements.

A soft sensor can forecast pH and EC between lab samples, helping growers detect drift earlier and adjust irrigation/fertigation decisions.

02 Project Goal

- Predict future root-zone pH and EC from the latest lab measurement, forecasted climate, planned irrigation/fertigation and crop-state data.
- Estimate the $t_0 \rightarrow t_1$ change so growers can test a planned irrigation/fertigation strategy, see expected future pH/EC values, and adjust before conditions drift.

03 Data

16,682

10-min rows

109

pH / EC labels

20+

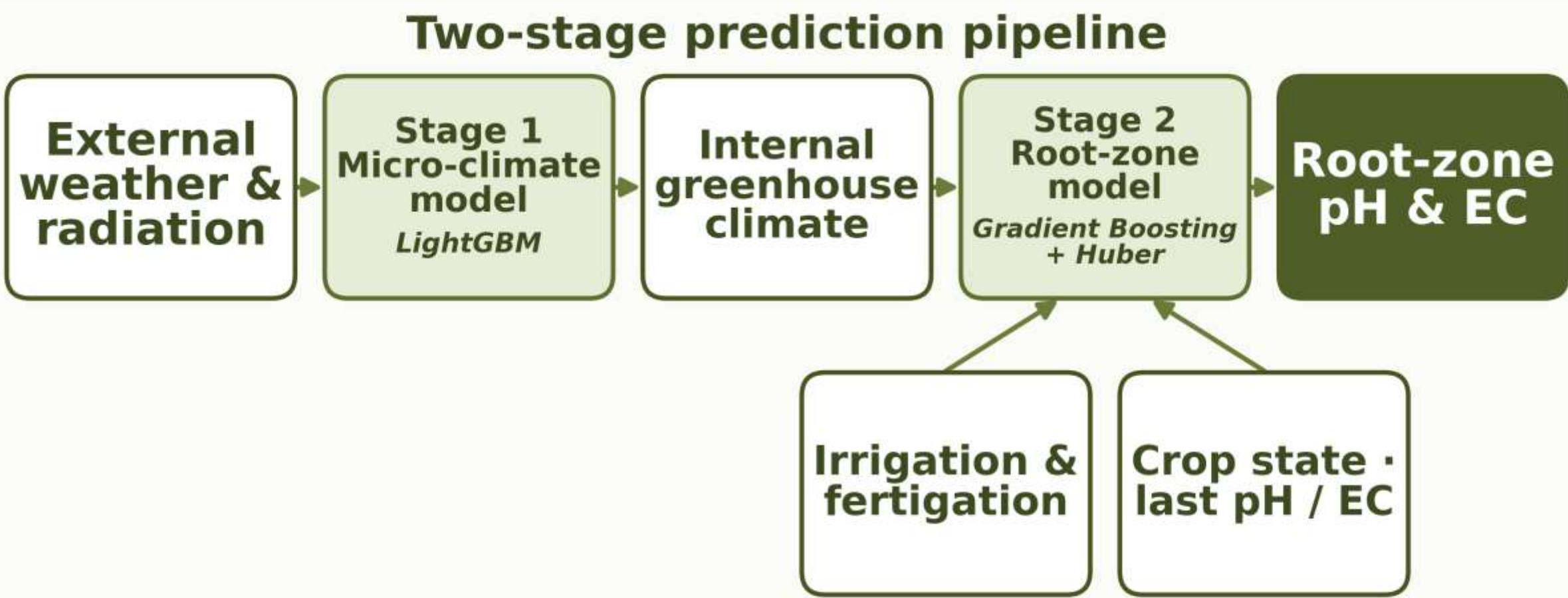
input features

- Unified 10-minute master dataset · May 29 – Sep 21, 2025.
- External weather & solar radiation.
- Greenhouse micro-climate.
- Irrigation & fertigation.
- Crop-state features.
- Manual laboratory pH & EC measurements.

04 Method

Two-stage prediction pipeline:

1. Forecast internal greenhouse micro-climate from external weather and radiation.
2. Predict $t_0 \rightarrow t_1$ root-zone pH/EC change using climate, crop state and planned events.



05 Algorithms & Model Choice

Hybrid model choice: nonlinear learning + robust modeling of rare extreme intervals.

Gradient Boosting

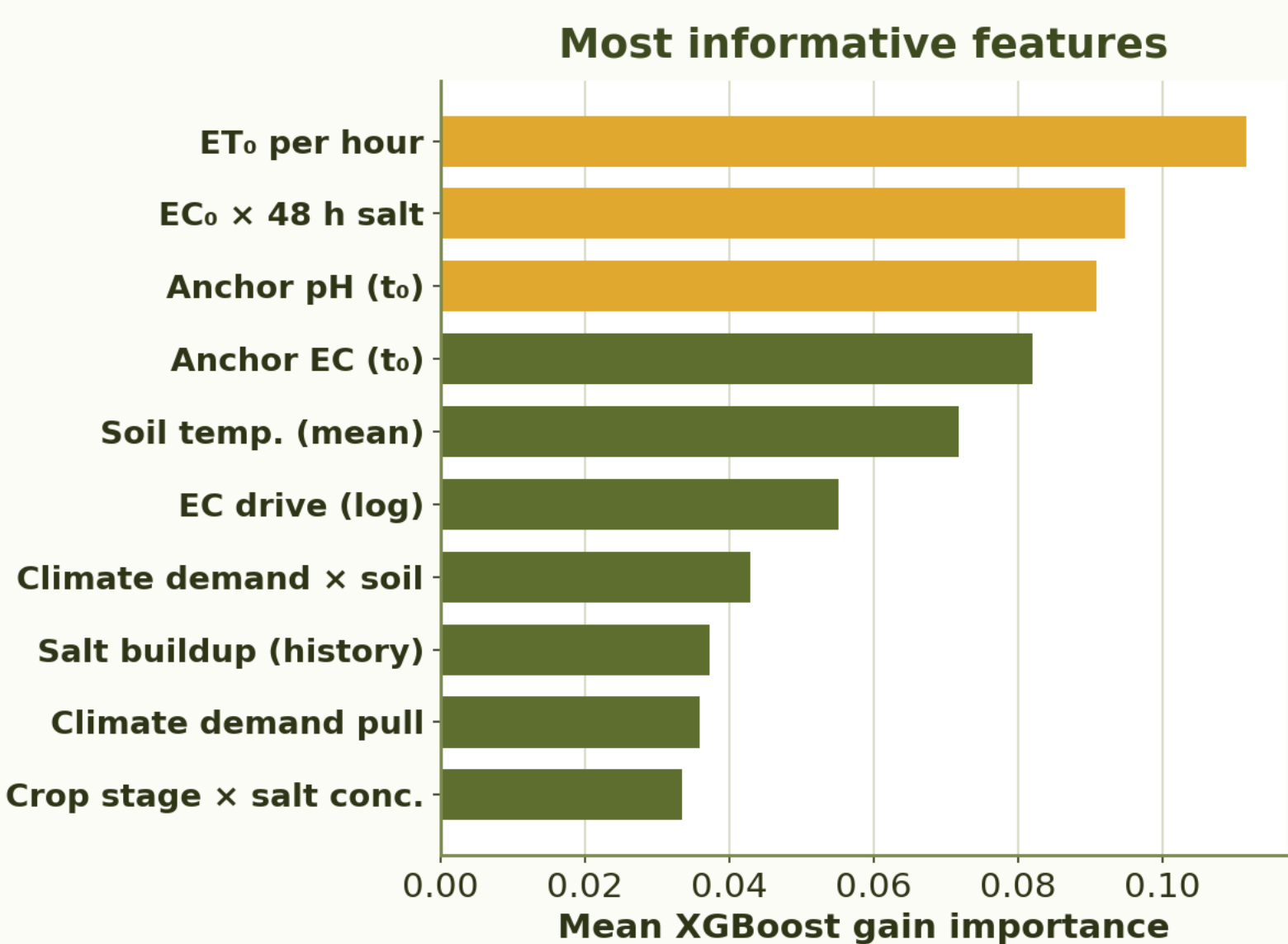
Main nonlinear model for tabular greenhouse data.

Captures interactions between climate, irrigation, fertigation, crop state and baseline pH/EC.

Huber Regressor

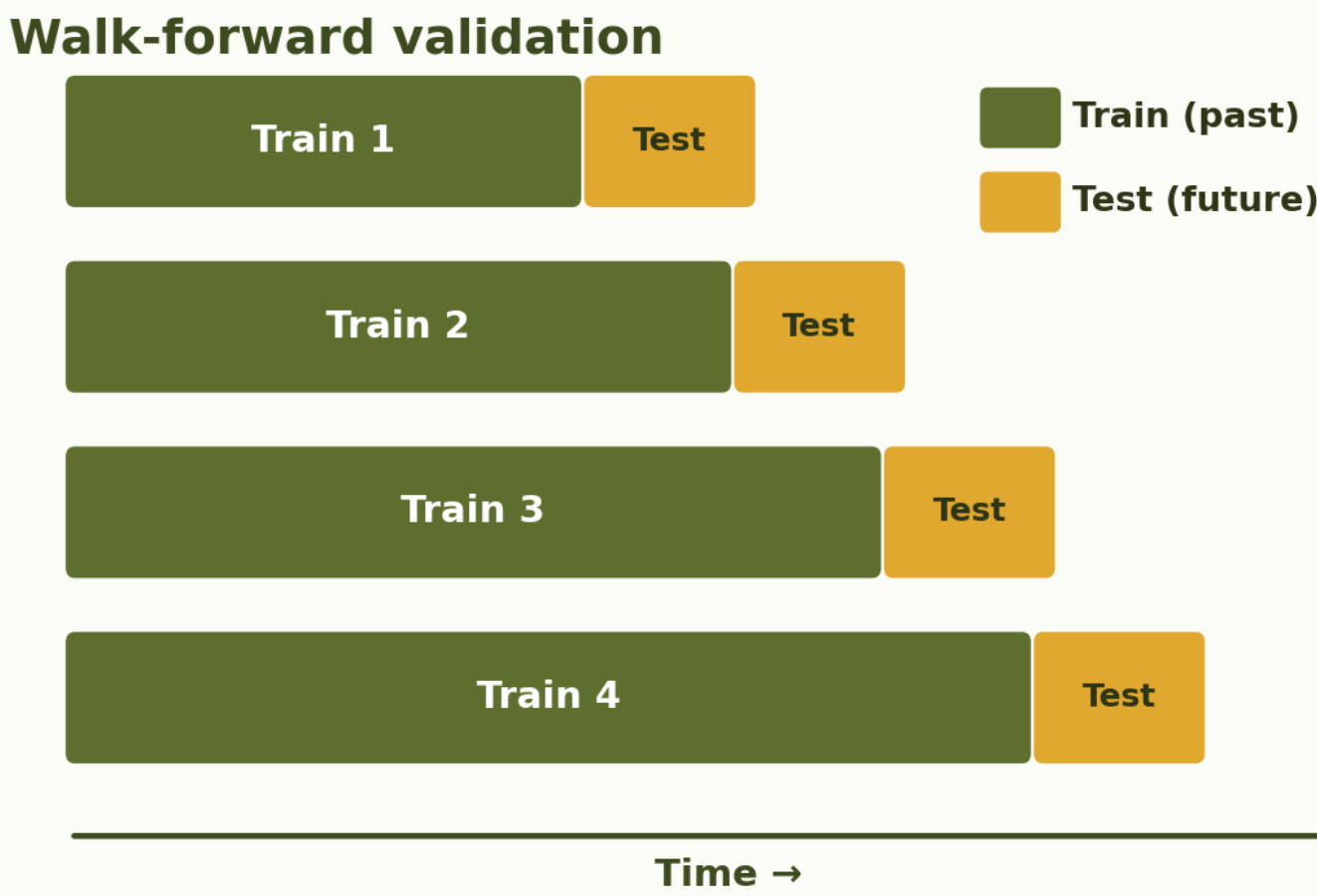
Robust companion model for sharp pH/EC shifts.

Reduces sensitivity to outliers caused by acid injection, fertigation and salinity events.



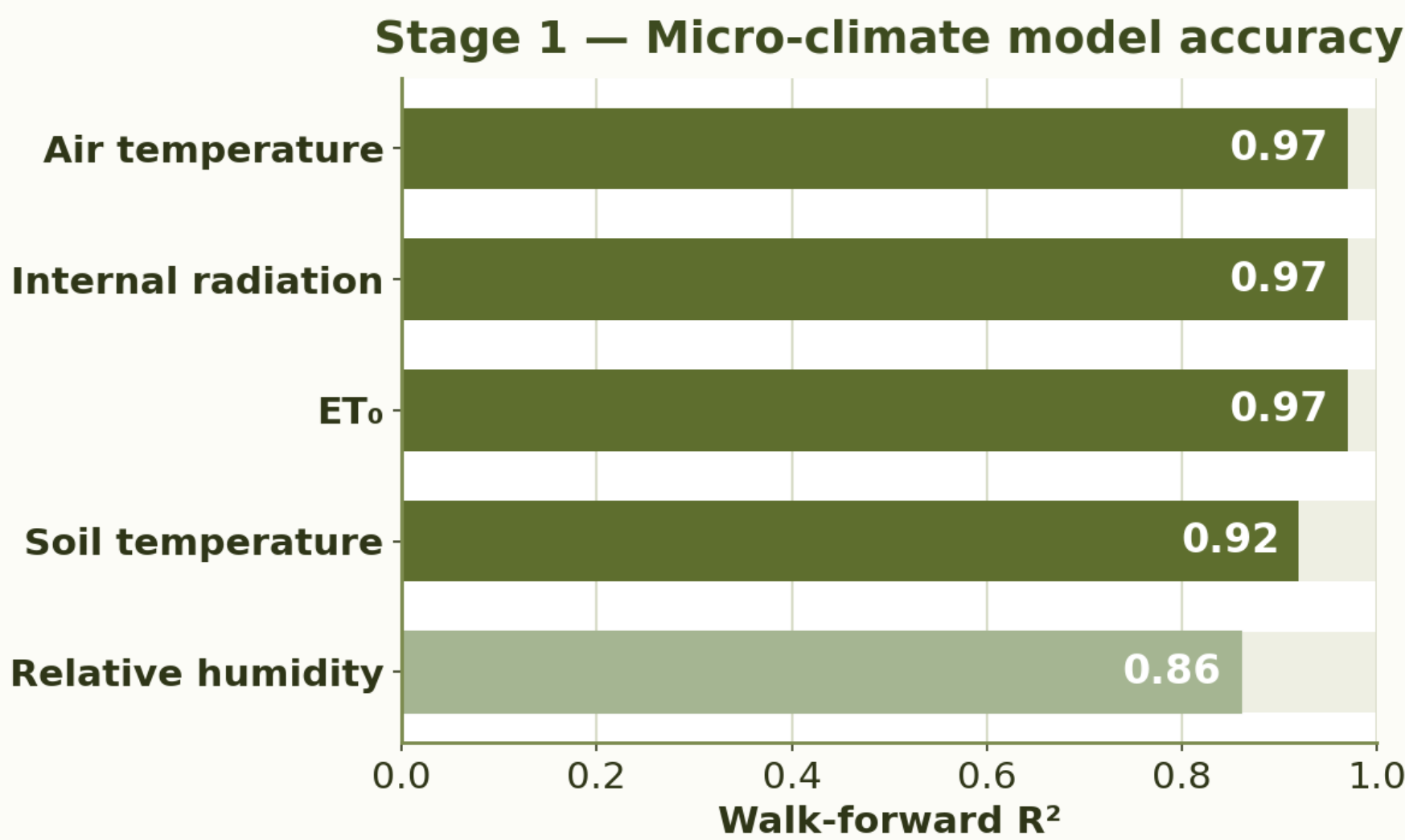
06 Validation Strategy

- Walk-forward validation matches deployment: future pH/EC is predicted from past labels, forecasted climate and planned irrigation/fertigation.
- Future pH/EC labels are never used for training until after that sample becomes available.



07 Micro-Climate Model

- Predicts internal greenhouse climate from external weather and radiation.
- Generates continuous climate inputs - temperature, RH, radiation, ET₀ and soil temperature - for the root-zone model.



08 Root-zone Model Results - Walk-forward Test

0.33 pH MAE

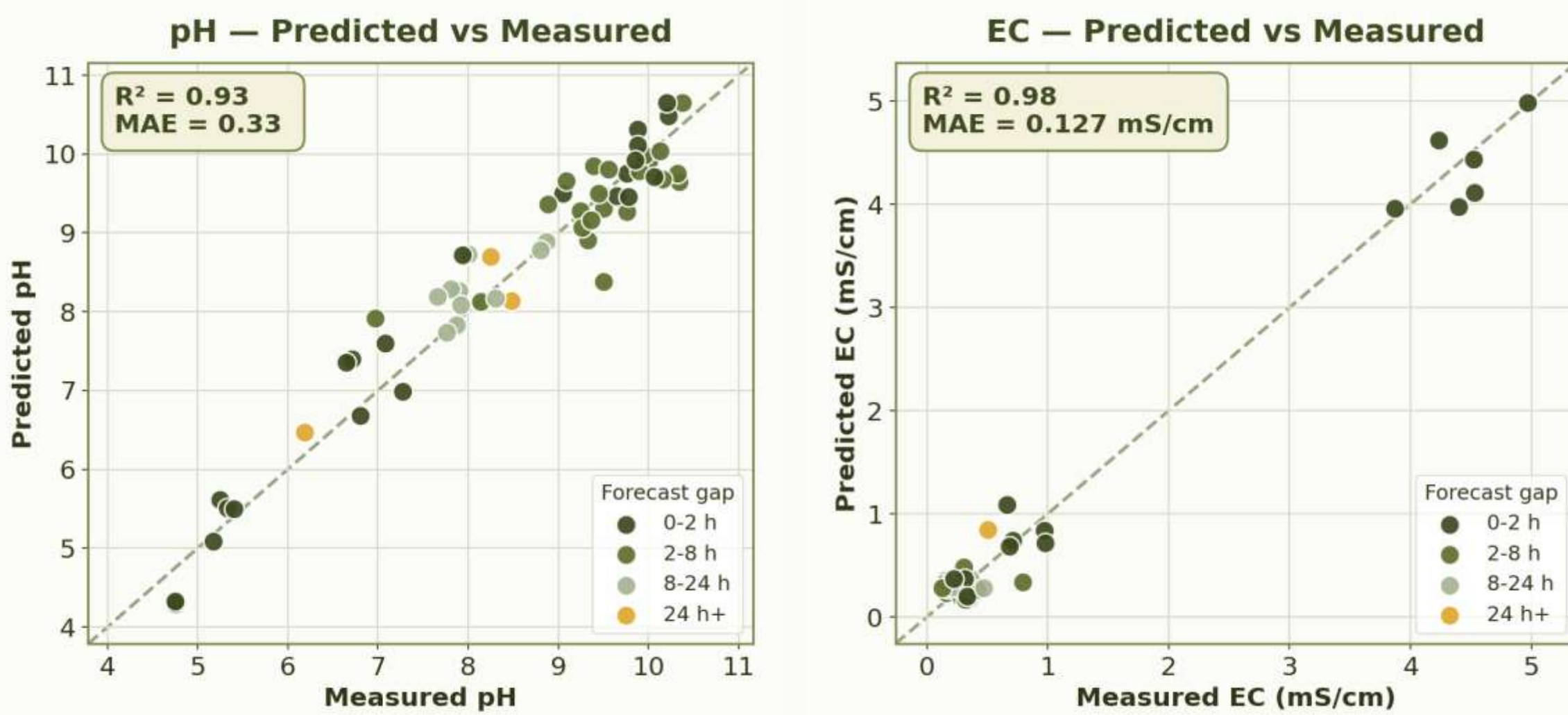
0.93 pH R²

+62% pH gain

0.127 EC MAE

0.98 EC R²

+35% EC gain



09 Conclusions

- The model substantially improves over the last-measurement baseline (+62% pH, +35% EC MAE gain), making it useful for early drift detection.
- In practice, the forecast can help growers test planned irrigation/fertigation actions before applying them.
- The system is ready as a decision-support prototype; broader validation is needed before autonomous control.

10 Challenges & Limitations

- Only 109 manual pH/EC labels against 16,682 dense sensor records.
- No continuous ground-truth pH/EC between lab samples.
- Irregular sampling intervals complicate interval modeling.
- Higher uncertainty in rare or extreme fertigation / salinity cases.
- Limited generalization: single season and one crop.